Cyber Risk Auditing Tool for Hybrid Work Enviroments

A ligthweight decision-support system for SMEs in hybryd work enviroments

Yessenia Sandoval (001070121) –BSc (Hons) Computer Science (Cybersecurity)
University of Greenwich, School of Computing and Mathematical Science
Email: ys6191n@gre.ac.uk

UNIVERSITY OF GREENWICH

INTRODUCTION

This project presents a web-based cyber risk auditing tool designed for hybrid work environments. It simplifies complex frameworks such as NIST and ISO 27001 into an accessible format for SMEs. The system enables users to assess cybersecurity risks and generate structured reports without requiring technical expertise.

WHY?

Cyber threats are growing. Hybrid work has expanded attack surfaces, and the average data breach now costs **$4.45 million** (IBM, 2024). SMEs face the greatest risk yet NIST and ISO/IEC 27001 are too complex without specialist expertise.

AIM / OBJECTIVES

AIM

To design, develop and evaluate a web-based cyber risk auditing tool for SMEs operating in hybrid work environments.

OBJECTIVES

1. Review cybersecurity risks in hybrid environments
2. Define requirements based on NIST and ISO/IEC 27001
3. Develop a web-based prototype for SMEs
4. Evaluate using simulated organisational scenarios

BACKGROUND

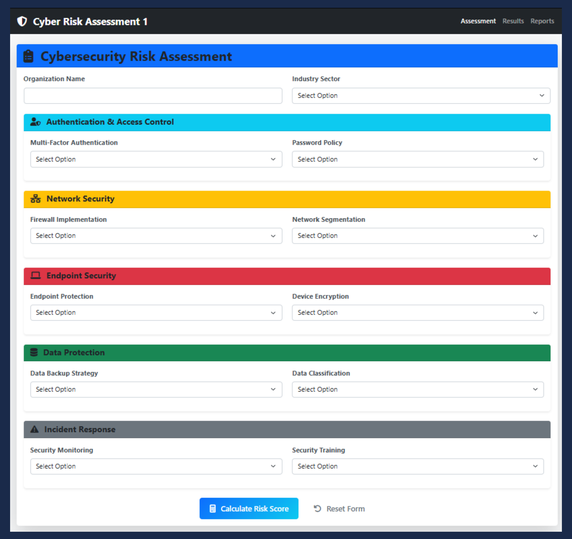
- NIST CSF and ISO/IEC 27001 guide risk management
- Complexity limits SME adoption
- Existing tools lack usability for hybrid work

APPROACH

- **DESIGN:** Defined system requirements based on NIST CSF and ISO/IEC 27001
- **DEVELOPMENT:** Implemented a web-based application with a risk scoring engine and user interface
- **EVALUATION:** Tested using simulated organisational scenarios to assess accuracy and usability

ARCHITECTURE

- Web-based cyber risk auditing tool
- Modular structure: frontend, backend, database
- Risk calculation engine processes user inputs
- Outputs: risk scores, analytics, reports



System workflow: inputs → risk calculation → outputs.

RISK MODEL

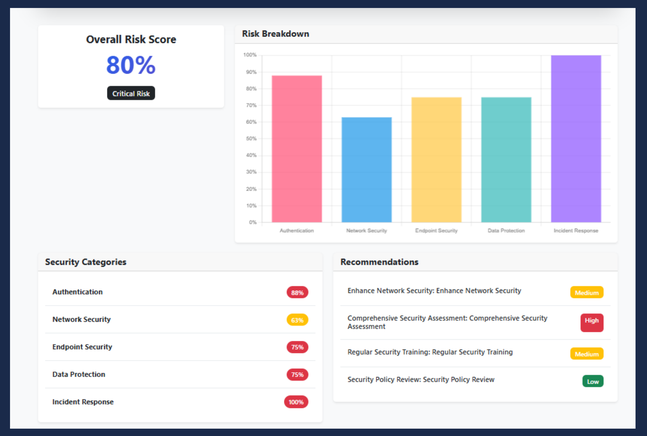
- Five domains: Authentication, Network, Endpoint, Data, Incident Response
- Weighted scoring model (20%) reflects domain importance
- Each control scored from 1 (Poor) to 4 (Excellent)
- Overall score calculated using weighted average

Risk levels: Low (0–25%), Medium (26–50%), High (51–75%), Critical (76–100%)

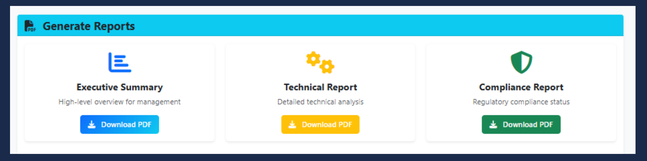
RESULTS AND EVALUATION

- Quantitative risk scoring
- Visual domain breakdown
- Automated recommendations
- Multi-format PDF reporting

Automated PDF Reports



Example output showing overall risk score and breakdown.



Example output showing overall reporting style.

KEY INSIGTHS

- Simplifying frameworks improves SME accessibility
- Lightweight tools support practical decision-making
- Hybrid work requires adaptive risk assessment
- Visual feedback enhances user understanding

LIMITATION AND FUTURE WORK

- Based on simulated scenarios (not real-world data)
- Simplified scoring model
- No real-time threat intelligence

Future Work

- Integrate real-time threat data
- Improve scoring model accuracy
- Expand testing with real organisations
- Enhance usability and scalability